METRIC

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PERFORMANCE SPECIFICATION

COOLANT FLUID, HYDROLYTICALLY STABLE, DIELECTRIC

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the characteristics and provides the requirements for a -54 °C to 200 °C synthetic hydrocarbon dielectric coolant fluid for use in electronic applications. This fluid is identified by NATO Code S-1748 (see 6.8).

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791 - Testing Method of Lubricants, Liquid Fuels, and Related Products

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE FORMS MANAGEMENT PROGRAM

DD Form 1718 - Certification of Qualified Products

(Copies of this document are available online at https://www.esd.whs.mil/DD.)

DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6 - Provisions Governing Qualification

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

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ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D664	-	Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration
ASTM D877/ D877M	-	Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes
ASTM D1169	-	Standard Test Method for Specific Resistance (Resistivity) of Electrical Insulating Liquids
ASTM D2532	-	Standard Test Method for Viscosity and Viscosity Change After Standing at Low Temperature of Aircraft Turbine Lubricants
ASTM D4289	-	Standard Test Method for Elastomer Compatibility of Lubricating Greases and Fluids
ASTM D4636	-	Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oils
ASTM D6304	-	Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration
ASTM D7647	-	Standard Test Method for Automatic Particle Counting of Lubricating and Hydraulic Fluids Using Dilution Techniques to Eliminate the Contribution of Water and Interfering Soft Particles by Light Extinction

(Copies of these documents are available from https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 2230	-	Rubber Products – Guidelines for Storage
ISO 4406	-	Hydraulic Fluid Power – Fluids – Method for coding the level of contamination by solid particles
ISO 11171	-	Hydraulic fluid power – Calibration of automatic particle counters for liquids
ISO 11218	-	Aerospace – Cleanliness classification for hydraulic fluids

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ISO 11500 - Hydraulic fluid power – Determination of the particulate contamination level of a liquid sample by automatic particle counting using the light-extinction principle

(Copies of these documents are available from https://www.iso.org.)

AEROSPACE INDUSTRIES ASSOCIATION / NATIONAL AEROSPACE STANDARDS (AIA/NAS)

AIA/NAS1638* - Cleanliness Requirements of Parts Used In Hydraulic Systems

* Indicates document is inactive for new design after 30 May 2001.

(Copies of this document are available from https://www.aia-aerospace.org.)

SAE INTERNATIONAL

SAE AMS3217/2 - Test Slabs, Acrylonitrile Butadiene (NBR-L) Low Acrylonitrile, 65-75

SAE AMS3400 - Fluid, Reference for Testing Polyalphaolefin (PAO) Resistant Material

SAE AS4059 - Contamination Classification for Hydraulic Fluids

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The dielectric coolant fluids furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Changes shall not be permitted in the formulation of an approved product unless specific, written approval of the qualifying activity is obtained.

3.2 Materials. The materials used in formulating this dielectric coolant fluid shall consist of synthetic, hydrocarbon base oil with suitable additive(s). A hydrogenated, polyalphaolefin base fluid with an additive to inhibit oxidation is recommended but not mandatory. The required properties shall be as specified in Table I and sections 3.3 through 3.4. Additional data is included in Table IV to serve as user information, which shall be referred to as typical (inherent)

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for a finished dielectric coolant fluid and shall not be a requirement levied upon the supplier (see 6.1.2).

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs.

3.3 Properties of finished fluid. The properties of the finished fluid shall be as specified in Table I and sections 3.3 through 3.4 when tested as specified in section 4.4.

3.3.1 Swelling of synthetic rubber. When tested as specified in section 4.4.4, the volume change of the standard synthetic rubber NBR-L (SAE AMS3217/2) by the dielectric coolant fluid shall be within the range of 0.0 to +10 percent.

3.3.2 Compatibility. The dielectric coolant fluid shall be compatible with other fluids of the same type, covered by this specification and identified in QPL-87252, when tested in accordance with section 4.4.2.

3.3.3 Low-temperature viscosity stability. The kinematic viscosity shall be tested, as specified in Table 1, at a temperature of -54 °C ± 1 °C. The viscosity at 1.5 hours, 3 hours, and 72 hours shall not exceed 1300 mm^2/s.

3.3.4 Storage stability. The fully blended product shall be clear and transparent, with no separation of ingredients or evidence of crystallization, and shall conform to all requirements of this specification after 12 months of storage.

3.3.5 High-temperature stability. When tested in accordance with section 4.4.3, the change in fluid viscosity at 40 °C shall not exceed 5 percent. The acid number of the fluid shall not increase by more than 0.25 mg KOH/g, and there shall be no formation of precipitate or insoluble material.

3.4 Workmanship. The workmanship shall be in accordance (IAW) with high-grade, commercial practices covering this type of material. The finished fluid shall be homogeneous and free from suspended matter, grit, or other adulteration.

3.5 Toxic products and formulations. The dielectric coolant fluid shall have no adverse effect on the health of personnel when used for its intended purpose.

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TABLE I. Properties of finished fluid.

Property	Qualification	Conformance	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Acid number, mg KOH/g (max)	X	X	0.20		D664	
Compatibility	X		(1)			4.4.2
Corrosiveness and oxidation stability (2) (168 hours @ 121 °C ± 1 °C)	X	X			D4636	
Change in acid number, mg KOH/g (max)	X	X	0.5			
Percent change in viscosity @ 40 °C (max)	X	X	5			
Metal specimen weight change (3), mg/cm²						
1010 Steel	X	X	± 0.2			
Aluminum	X	X	± 0.2			
Magnesium	X	X	± 0.2			
Cadmium plated steel (4)	X	X	± 0.2			
Copper (5)	X	X	± 0.4			
Separation of insoluble materials or gumming of the fluid	X	X	None			
Dielectric strength, kV (min)	X	X	35		D877 /D877M	
Flash point, °C (min)	X	X	150		D92	
Fire point, °C (min)	X		160		D92	

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TABLE I. Properties of finished fluid - Continued.

Property	Qualification	Conformance	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
High temperature stability (200 °C ± 2 °C for 100 hrs)	X	X				4.4.3
Change in acid number, mg KOH/g (max)	X	X	0.25			
Percent change in viscosity @ 40 °C (max)	X	X	5			
Formation of Precipitate or Insoluble Material	X	X	None			
Particulate contamination (6) : One of the following shall be measured.						
Particle count, AS4059 Contamination Class (7) (max)	X	X	5 (8)	3012 (9) (10)	(11500)	4.4.1
Particle counting, individual channel counts & ISO Code (max)	X	X	16/14/11(11)		D7647 (9) (11500)	4.4.1
Resistivity @ 25°C, ohm-cm (min)	X	X	1.0×10^{10}		D1169	
Rubber swell, standard synthetic rubber, NBR-L(12), percent change in volume	X		0.0-10.0		D4289 (13)	4.4.4

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TABLE I. Properties of finished fluid - Continued.

Property	Qualification	Conformance	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Storage stability (24 °C ±3 °C for 12 months)	X		Section 3.3.4 (14)			4.4.5
Viscosity, mm²/sec						
@ -54 °C ±1 °C @ 1.5 hrs (max)	X	X	1300		D2532	
@ -54 °C±1 °C @ 3 hrs (max)	X	X	1300		D2532	
@ -54 °C±1 °C @ 72 hrs (max)	X	X	1300		D2532	
@ -40 °C (max)	X	X	300		D445	
@ 40 °C (min)	X	X	5.0		D445	
@ 100 °C (min)	X	X	1.65		D445	
Water, mg/kg (max)	X	X	50		D6304 (15)	
Workmanship	X	X	(16)			3.4

Notes:

1.	Compatible with other fluids of the same type, covered and when tested in accordance with this specification. None of the mixtures shall show any signs of sediment, turbidity, or crystallization. After samples are stored at -54 °C for 2 hours, a slight turbidity during this time, which later disappears, will be permitted in the samples.
2.	Test conditions: 168 hours at 121 °C. Use Alternative Procedure 2. Use heptane or acetone as the solvent.
3.	There shall be no pitting, etching, or visible corrosion on the surface of the metals viewed under magnification of 20 diameters.
4.	A slight discoloration of the cadmium shall be permitted.
5.	Any corrosion produced on the surface of the copper shall be no greater than No. 3a of the ASTM D130 copper strip corrosion standard.

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TABLE I. Properties of finished fluid - Continued.

Notes:	
6.	Particulate contamination shall be measured as either particle count, AS4059 Contamination Class or Particle counting, individual channel counts & ISO Code (ISO 4406). In alignment with industry practices, particle counting, individual channel counts, with corresponding ISO Code, is the preferred reporting method.
7.	The container shall be thoroughly shaken immediately prior to withdrawing the sample. A determination shall be made on 100 mL portion of the sample
8.	See Table II for particle count size ranges.
9.	Referee method.
10.	Hexane shall be used instead of ozone depleting solvents.
11.	Results shall be reported for particle size, channel count, and ISO code in same format and criteria as laid out in Table III.
12.	Test conditions: 168 hours at 70 °C.
13.	When checking elastomers for conformance to the specification as noted in D4289 annex "Physical Property Data for Reference Elastomers" use SAE AMS3400 as the reference fluid.
14.	The fully blended product shall be clear and transparent, with no separation of ingredients or evidence of crystallization, and shall conform to all requirements of this specification.
15.	Use Procedure A (Direct Injection).
16.	The finished fluid shall be homogeneous and free from suspended matter, dirt, or other foreign material.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of testing to all the requirements specified in sections 3.3 through 3.4 and Table I. When required by the qualifying activity, additional evaluations may be required on candidate formulations.

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4.2.1 Qualification process. At the initiation of the qualification process, prospective suppliers will forward a written request via email for such action to the qualifying activity, Air Force Petroleum Office, AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 (afpet.ptps@us.af.mil). This request will contain general information on the proposed candidate material. The qualifying activity will reply in writing outlining the required information to be submitted, the governing regulations for qualified products, and the cost of testing. The information that will be submitted includes:

a. Letter stating the company will adhere to the appropriate provisions of the SD-6 on company letterhead.

b. Complete formulation of the candidate product, including chemical composition in weight percent, the manufacturer, the commercial trade name and CAS number, the purities, and use of each component.

c. Safety Data Sheet (SDS) for each component and final product. The SDS will not be older than 5 years.

d. Certified laboratory test data from the manufacturer, a commercial laboratory, or combination thereof showing quantitative results of all qualification tests required by this specification as identified in Table I.

e. Verification that the System for Award Management (SAM) (CAGE Codes for the corporate office and each plant) is current. This can be viewed at https://sam.gov/.

f. Identification of the manufacturing site of the specific batch of test sample to be submitted.

g. A completed DD Form 1718.

4.2.2 Qualification inspection sample. A qualification sample shall consist of a one-gallon container of dielectric coolant fluid. The qualifying activity will request manufacturer and third-party data prior to authorizing submission of the qualification sample.

4.2.3 Retention of qualification. To retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer will be requested to verify by certification (DD Form 1718) to the qualifying activity that its product(s) comply with the requirements of this specification. The manufacturer shall also provide a recent Certificate of Analysis (COA), to include all conformance inspection requirements (see 4.3), a current SDS, and formulation. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in 2-1/2 year intervals from the date of original qualification and will be initiated by the manufacturer. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

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4.2.4 Requalification. Requalification shall be required every five (5) years or when any change is made in the source of manufacture, purity, or composition of the fluid's base stock or additives. A minor change in the fluid formulation such as a new supplier of stock material might not need requalification testing, but only after notification to, and approval by the qualifying activity (see 6.3). The documentation required for requalification are the same as requirements stated in section 4.2.1 and shall include product formulation, safety data sheet (SDS) for each component and final product, a COA showing compliance to all of the qualification requirements specified in Table I and a one (1) gallon requalification sample from a recent production run within the previous 60 days. The SDS will not be older than five (5) years. The qualifying activity may, at its discretion, waive complete requalification or may require only partial requalification testing to determine the significance and acceptability of the proposed formulation change.

4.3 Conformance inspection. The conformance inspection shall consist of all the tests in Table I and sections 3.3 through 3.4, except for rubber swell, compatibility, and storage stability.

4.4 Methods of inspection. Tests shall be performed in accordance with the applicable methods specified in Table I and section 4. Physical and chemical values, specified in section 3, apply to the arithmetic average of the determinations made on the samples for those values which fall within any stated repeatability or reproducibility limits of the applicable test method.

4.4.1 Particle count. Particle count shall be determined as follows:

a. An automatic particle counter employing either the light extinction principle or optical imaging and meeting the channel, particle size range, and calibration requirements specified in Table II or Table III shall be used (use the operating instructions for the applicable instrument).

b. Mix the sample thoroughly immediately prior to withdrawing the sample.

TABLE II. Particle Count, SAE AS4059/ISO 11218 Contamination Level 5.

Particle Size Ranges, micrometers (µm) (1) AIA/NAS 1638 (2)		Maximum number of particles per 100 mL
SAE AS4059	SAE AS4059	DIFFERENTIAL
5-15	6-14(c)	8000
16-25	15-21(c)	1425
26-50	22-38(c)	253
51-100	39-70(c)	45
>100	>70(c)	8

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TABLE II. Particle Count, SAE AS4059/ISO 11218 Contamination Level 5 - Continued.

SAE AS4059	SAE AS4059	CUMULATIVE
>5	>6(c)	9731
>15	>14(c)	1731
>25	>21(c)	306
>50	>38(c)	53
>100	>70(c)	8
Notes:		
1. The symbol (c) is used to designate the automatic particle counter was calibrated to ISO 11171.		
2. AIA/NAS 1638 - Cleanliness Requirements of Parts Used in Hydraulic Systems is obsolete and replaced with SAE AS4059. Referenced for transition purposes only.		

TABLE III. Particle Count, ISO 4406 (SAE AS4059/ISO 11218 Contamination Level 5 Basis).

Particle Size Channels, micrometers (µm) (1)		
Size	Counts per mL	ISO Code (3)
≥ 4 µm(c)	(2)	Max 16
≥ 6 µm(c)	(2)	Max 14
≥ 14 µm(c)	(2)	Max 11
≥ 21 µm(c)	(2)	(2)
≥ 38 µm(c)	(2)	(2)
≥ 50 µm(c)	(2)	(2)
≥ 70 µm(c)	(2)	(2)

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TABLE III. Particle Count, ISO 4406 (SAE AS4059/ISO 11218 Contamination Level 5 Basis) - Continued.

Notes:
1. The symbol (c) is used to designate the automatic particle counter was calibrated to ISO 11171.
2. Report.
3. Per Table 1 “Allocation of scale numbers” of ISO 4406.

4.4.2 Compatibility. Samples of candidate dielectric coolant fluid in amounts of 20 mL, 100 mL, and 180 mL shall be mixed with samples from each of the fluids currently approved under this specification and listed on QPL-87252. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://quicksearch.dla.mil via a search using 87252 for the document number and selecting the QPL-87252 document. Total volume of each mixture shall be 200 mL. Mixtures shall be prepared in 250 mL stoppered flasks. The flasks shall be thoroughly shaken and then stored in an oven at 135 °C for 2 hours. At the end of this time, none of the mixtures shall show any signs of sediment, turbidity, or crystallization. The samples shall then be stored at -54 °C for a period of 2 hours. Slight turbidity during this time, which later disappears, will be permitted in the samples.

4.4.3 High-temperature stability.

4.4.3.1 Apparatus. Thermal-stability characteristics of the coolant fluid shall be determined using the following apparatus: a 50 mL round-bottom flask or test tube (25 mm x 50 cm) with a 24/40 taper (Ace Glass Inc., Vineland, NJ 08360 or equivalent); a 75° angle “Y” side arm adapter with standard 24/40 taper (Ace Glass 5040-10 or equivalent); borosilicate glass gas inlet tube (Koehler K351-0-3 or equivalent); PTFE thermocouple adapter, 24/40, ¼” probe (ChemGlass CG-1048-A-05 or equivalent); aluminum foil collar (if necessary); 6-mm Outside Diameter (OD) stainless steel tubing, type 304, 2.5 cm in length ; and a high-temperature bath capable of maintaining a temperature of 200 °C ± 2 °C.

4.4.3.2 Procedure. The following procedures shall be followed for determining high-temperature stability:

a. Place a 2.5-cm length of 6mm OD stainless steel tubing, type 304, into a 50 mL round-bottom flask or 25 mm x 50 cm test tube.

b. Pour 25 mL of candidate fluid into the flask or test tube.

c. Fit the flask or test tube with a 75° angle adapter.

d. Slide the PTFE thermocouple adapter onto the gas inlet tube and insert the gas inlet tube into the flask or test tube assembly, keeping the tip of the gas inlet tube above the

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surface of the fluid. Adjust the height of the glass inlet tube so that the tip will be approximately 1.0 cm above the surface level of the heat-expanded fluid.

e. Place the flask or test tube assembly into a high-temperature bath preheated to 200 °C ± 2 °C. If necessary, use a collar made of aluminum foil on the top of the flask or test tube to ensure the apparatus is stable and centered in the bath.

f. Ensure the dry nitrogen (lamp grade) source is at a flow rate of 1-liter-per hour, then connect it to the gas inlet tube on the apparatus.

g. Heat the fluid for 100 hours at a temperature of 200 °C ± 2 °C while maintaining a 1-liter-per hour flow of dry nitrogen (lamp grade) on the surface of the fluid via the gas inlet tube positioned within 1.0 cm above the level of the fluid.

4.4.3.3 Evaluation. Upon completion of the test, the fluid shall meet the requirements of Table I.

4.4.4 Rubber swell. Rubber swell shall be performed IAW ASTM D4289 with the following modifications:

a. Use the standard synthetic rubber, NBR-L meeting the latest revision of SAE AMS3217/2. Store in the dark, in an inert atmosphere, below 65% relative humidity, away from direct heat sources, and between 15 °C to 25 °C (59 °F to 77 °F) IAW ISO 2230.

b. Check elastomer for conformance to SAE AMS3217/2, latest revision, at least every 6 months from the date of manufacture using SAE AMS3400 as the reference fluid.

c. Age the specimens by storing the container in a convection oven at 70 °C ± 2.5 °C for 168 hrs ± 0.5 hrs.

4.4.5 Storage stability.

4.4.5.1 Apparatus. Storage stability characteristics of the dielectric coolant fluid shall be determined using the following test apparatus: A storage chamber that is able to maintain a constant-temperature of 24 °C ± 3 °C (75 °F ± 5 °F) and is free of vibration (such as caused by heavy traffic). A 3.8 L (1 gal) clear-glass wide-mouth jar with cap.

4.4.5.2 Procedure. The following procedures shall be followed for determining storage stability:

a. Clean the jar and cap by scrubbing with scouring powder and rinsing with tap water. Complete the rinsing with distilled water and air dry the jar and cap (using an oven at 104 °C (220 °F) if desired).

b. Line the cap with aluminum foil.

c. Pour approximately 3.8 L of the sample into the jar, and replace and tighten the cap.

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d. Wrap aluminum foil around the jar to exclude all light from the sample.

e. Identify the contents of the glass jar (including type of sample, date stored, and any other useful information).

f. Store the sample for 12 months (one year) in the storage chamber at 24 °C ± 3 °C (75 °F ± 5 °F).

g. At the end of the 12-month period, remove the jar from the storage chamber and take off the aluminum wrapper, being careful not to shake or agitate the contents.

h. Visually examine the sample, and note the presence of cloudiness, sediment, suspended matter, discoloration, or other changes in homogeneity.

4.4.5.3 Evaluation. Upon completion of the test, the fluid shall meet the requirements of Table I.

4.6 Examination of filled containers. A random sample of filled containers from each packaged lot (see 6.5.2) shall be examined with regard to fill, closure, sealing, and leakage.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use as a heat-transfer medium in closed military systems and capable of functioning as a dielectric coolant fluid over the temperature range of -54 °C to 200 °C. Dielectric coolant fluid is used to cool military aircraft systems, including radar, aircraft avionics systems, missiles, and navigation systems.

6.1.1 Storage conditions. Prior to use in intended equipment, the product may be stored under conditions of covered or uncovered storage in geographic areas ranging in temperature from -54 °C to 50 °C.

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6.1.2 Additional data for typical properties of the finished coolant fluid. Additional data is included in Table IV to serve as user information which will be referred to as typical (inherent) and will not be a requirement levied upon the supplier.

TABLE IV. Typical properties of the finished dielectric coolant fluid.

Characteristic	Inherent Value
Specific Gravity	
@ 25 °C (77 °F)	0.789
@ 15.6 °C (60 °F)	0.800
Coefficient of thermal expansion (cm^3/cm^3 °C)	8.1×10^{-4}
Thermal conductivity (cal-cm/s-cm^2 °C)	
@ -40 °C (-40 °F)	4.5×10^{-4}
@ 0 °C (32 °F)	3.8×10^{-4}
@ 150 °C (302 °F)	3.0×10^{-4}
Specific Heat	
@ -40 °C (-40 °F)	0.48
@ 25 °C (77 °F)	0.53
@ 75 °C (167 °F)	0.55
@ 149 °C (300 °F)	0.65

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Quantity required.

c. Type and capacity of containers.

d. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, contract awards will be made only for such products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 87252 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433; e-mail: afpet.ptps@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://quicksearch.dla.mil.

6.3.1 Qualification information. It is understood that the material furnished under this specification subsequent to final approval will be of the same composition and will be equal to products on which approval was originally granted. In the event that fluid furnished under contract is found to deviate from the composition of the approval product, or that the product fails to perform satisfactorily, approval of such products may be subjected to immediate withdrawal from the QPL at the discretion of the approving activity.

6.3.2 Proceed authorization notification. Upon successful review of the submitted information, the Air Force Petroleum Office will issue authorization to ship the qualification sample(s) for testing.

6.3.3 Qualification samples. Upon receiving authorization from the Air Force Petroleum Office, qualification sample(s) may be shipped to the qualification activity, as detailed in the authorization letter. Sample(s) should be accompanied by an SDS and a copy of the authorization letter. Each sample should be plainly identified by securely attached durable tags or labels marked with the following information:

a. QUALIFICATION SAMPLE

b. Coolant Fluid, Hydrolytically Stable, Dielectric

c. MIL-PRF-87252

d. Name of Manufacturer

e. Product Code Number

f. Batch Number

g. Date of Manufacture

6.4 Toxicity. Questions pertinent to this effect should be referred by the contracting activity to the appropriate departmental medical service who will act as its advisor.

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6.5 Definitions.

6.5.1 Bulk lot. A bulk lot is defined as an indefinite quantity of homogeneous mixture of material offered for acceptance in a single isolated container or manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.5.2 Packaged lot. A packaged lot is defined as an indefinite number of 208-liter (55-gallon) drums or smaller unit packages of identical size and type offered for acceptance and filled with a homogeneous mixture of material manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume I, Shelf-Life Management Program: Program Administration; and Volume 2, Shelf-Life Management Program: Materiel Quality Control Storage Standards. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website https://www.shelflife.dla.mil/ .

6.7 Subject term (key word) listing.

Heat transfer fluid

Insulating fluid

Liquid coolant

Synthetic hydrocarbon

Synthetic rubber

6.8 International standardization agreement implementation. This specification implements:

a. Air Force Interoperability Council (AFIC) Air Standard Fuels, Lubricants and Gases Working Group (FLG) 4023 *Hydraulic Fluid, Petroleum (H-515) and Polyalphaolefin (H-537 and H-538)*.

b. Air Force Interoperability Council (AFIC) AIR STANDARD Fuels, Lubricants and Gases Working Group (FLG) 4024, *Interchangeability Chart of Standardized Aviation Fuels, Lubricants, and Associated Products*.

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c. NATO STANAG/AFLP 1135, *Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations*.

d. NATO STANAG/AFLP-3748 *Hydraulic Fluid, Petroleum (H-515), Polyalphaolefin (H-537, H-538) and Phosphate Ester Fluids (H-522, H-523, H-524)*.

When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.9 Conversion of metric units. Units of measure have been converted to the International System of Units (Metric) IAW IEEE/ASTM SI10 *American National Standard for Metric Practice*. If test results are obtained in units other than Metric or there is a requirement to report dual units, IEEE/ASTM SI10 or NIST SP 811 *Guide for the Use of the International System of Units (SI)* should be used to convert the units.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – AS
Air Force – 68

Preparing activity:
Air Force – 68
(Project 9150-2023-008)

Review activities:
Army – AT, MI
Navy – SH
Air Force – 20
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the ASSIST Online database at https://assist.dla.mil.

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